

§ 11.6 Lines & tangent lines in 3 space

Recall the equation of a line (through $\vec{r}_0$ with direction $\vec{v}$).

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

$$\text{i.e. } x = x_0 + ta, \quad y = y_0 + tb, \quad z = z_0 + tc$$

$$\text{where } \vec{r}_0 = (x_0, y_0, z_0) \text{ and } \vec{v} = (a, b, c)$$

Ex Find a parametric equation for the line through $(1, -1, 2)$ & $(3, 1, -2)$.

$$\vec{v} = (3-1, 1-(-1), -2-2) = (2, 2, -4) \quad \vec{r}_0 = (1, -1, 2)$$

$$\vec{r} = (1, -1, 2) + t(2, 2, -4) = (1+2t, -1+2t, 2-4t)$$

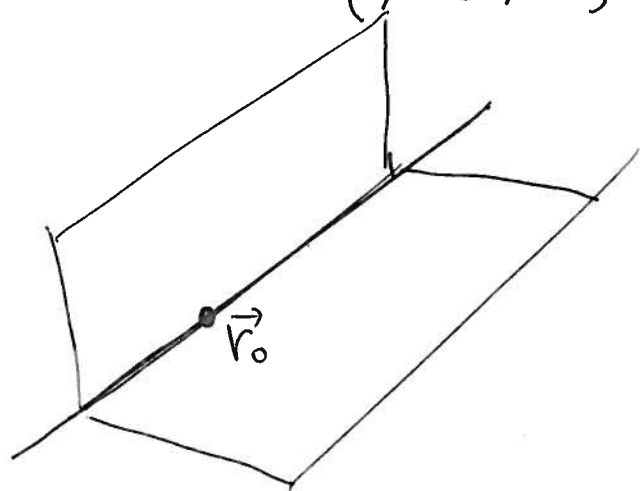
Another point of view:
$$\begin{cases} x = 1+2t \\ y = -1+2t \\ z = 2-4t \end{cases} \Rightarrow \begin{cases} t = \frac{x-1}{2} \\ t = \frac{y+1}{2} \\ t = \frac{z-2}{-4} \end{cases}$$

$$\text{so } \frac{x-1}{2} = \frac{y+1}{2} = \frac{z-2}{-4}$$

so the line is the intersection of 2 planes

$$\frac{x-1}{2} = \frac{y+1}{2} \quad \& \quad \frac{y+1}{2} = \frac{z-2}{-4}$$

Ex Find parametric eqns for the line given by $\begin{cases} x+y+z=1 \\ x-2y=3 \end{cases}$



$$\vec{n}_1 = (1, 1, 1)$$

$$\vec{n}_2 = (1, -2, 0)$$

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 1 \\ 1 & -2 & 0 \end{vmatrix}$$

$$= (2, 1, -3)$$

$$\vec{r}_0 = (3, 0, -2)$$

$$\vec{r} = (3, 0, -2) + (2, 1, -3) \cdot t = (3+2t, t, -3t)$$

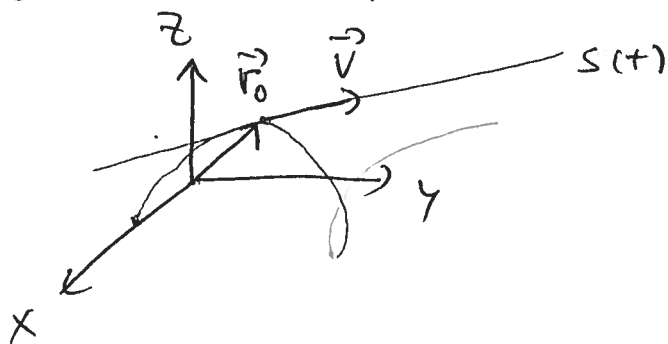
Ex Find the tangent line to $\vec{r}(t) = (\cos t, t, \sin t)$ for $t = \pi/2$

A: $\vec{r}'(t) = (-\sin t, 1, \cos t)$

$$\vec{r}_0 = \vec{r}(\pi/2) = (0, \pi/2, 1)$$

$$\vec{v} = \vec{r}'(\pi/2) = (-1, 1, 0)$$

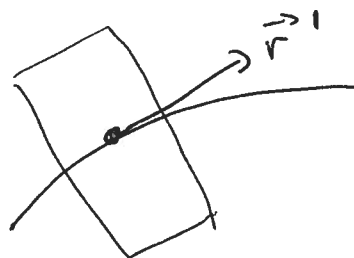
tg line is $s(t) = (0, \pi/2, 1) + t(-1, 1, 0) = (-t, \pi/2 + t, 1)$



Σ_x Find the plane perpendicular to $\vec{r}(t) = (t, t^2, t^3)$ at $t = -2$

A: $\vec{r}'(t) = (1, 2t, 3t^2)$ $\vec{r}(-2) = (-2, 4, -8)$

$\vec{r}'(-2) = (1, -4, 12)$



$x - 4y + 12z = D$

$-2 - 4 \cdot 4 + 12 \cdot (-8) = D$

$D = -114$

so $x - 4y + 12z = -114$

Σ_x Write parametric (and symmetric) equations

- for the line through $(1, -2, 3)$ & $(4, 5, 6)$
- for the line through $(-1, 3, -6)$ parallel to $(-2, 0, 5)$
- the line through $(-5, 7, -2)$ perpendicular to $(2, 1, -3)$ and to $(5, 4, -1)$
- the tangent line to $\begin{cases} x = 2t^2 \\ y = 4t \\ z = t^3 \end{cases}$ at $t = 1$

Σ_x Find the equation of the plane

- containing the lines $\begin{cases} (1, 0, 1) + t(2, -1, -1) \\ (3, -1, 0) + s(1, 0, 2) \end{cases}$
- containing the line $\begin{cases} x = 1 + 2t \\ y = -1 + 3t \\ z = 4 + t \end{cases}$ & the point $(1, -1, 5)$
- perpendicular to $\vec{r}(t) = (t \sin t, 3t, 2t \cos t)$ at $t = \pi/2$